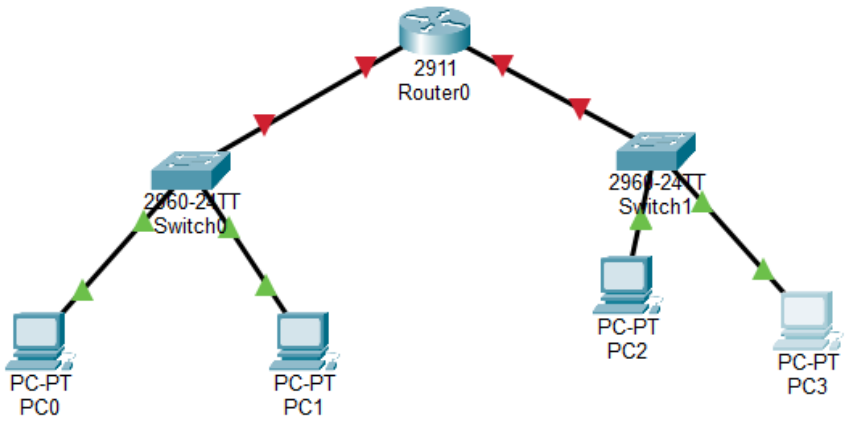
Project 01 | LAN & DHCP Troubleshooting

Screenshot evidence from five Cisco Packet Tracer incident investigations (INC-001 through INC-005).

Includes configuration baselines, reproduced faults, diagnostic commands, repairs, and verification results. Images are presented in the order supplied.

Topology and baseline

Evidence 01: Topology screenshot



INC-001 - Router interface down

Evidence 01: Baseline - both interfaces up

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Evidence 02: Failure - interface administratively down

```
Router#show ip interface b
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Evidence 03: Failure - ping unreachable

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Evidence 04: Repair - interface up/up

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Evidence 05: Verification - ping successful

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

INC-002 - Incorrect default gateway

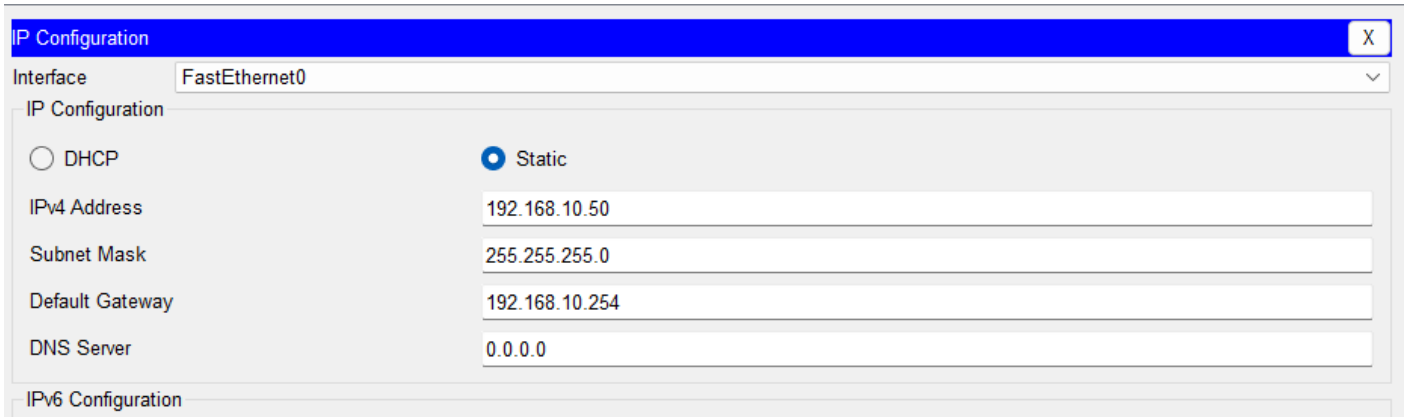
Evidence 01: Initial PC configuration without IPv4 address

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address.................: 0009.7C83.8681
   Link-local IPv6 Address.........: FE80::209:7CFF:FE83:8681
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 192.168.10.1
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-B6-A9-33-B6-00-09-7C-83-86-81
   DNS Servers.....................: ::
                                     0.0.0.0
```

Evidence 02: Incorrect default gateway configured



Evidence 03: Local ping succeeds

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Evidence 04: Remote ping fails

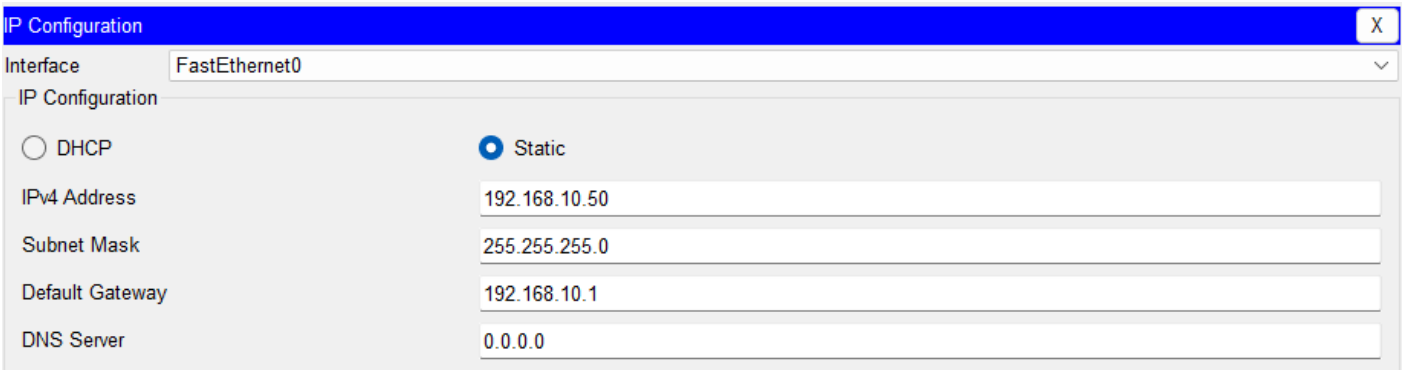
```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),
```

Evidence 05: Gateway corrected



Evidence 06: Remote ping succeeds

```
C:\>ping 192.168.20.1

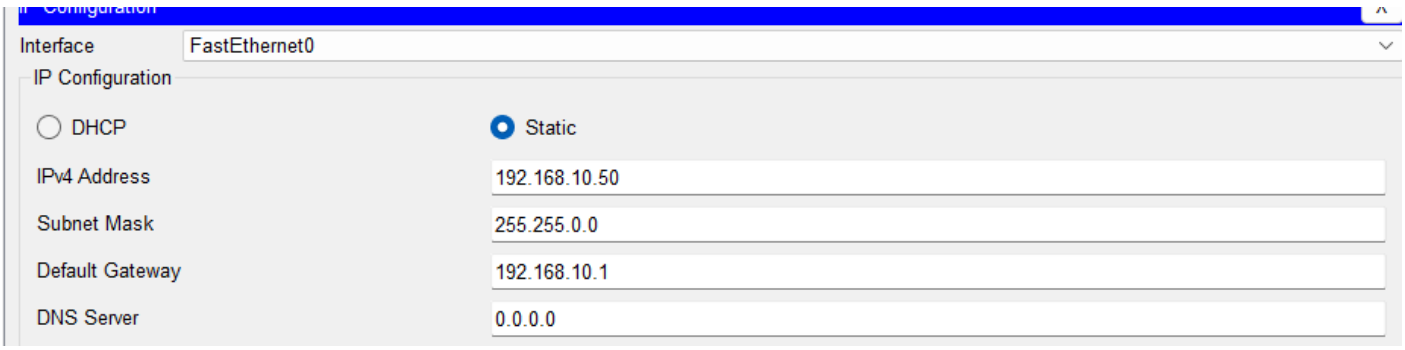
Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

INC-003 - Subnet mask and Proxy ARP

Evidence 01: Incorrect /16 subnet mask



Evidence 02: Ping succeeds despite wrong mask

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Evidence 03: ARP entry for remote address

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.20.1          0060.3e4b.3401        dynamic
```

Evidence 04: Proxy ARP enabled

```
Router#show ip interface g
Router#show ip interface gigabitEthernet 0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
```

Evidence 05: Proxy ARP disabled for diagnosis

```
Router# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#in
Router(config)#interface g
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no ip pro
Router(config-if)#no ip proxy-arp
Router(config-if)#end
Router#
%SYS-5-CONFIG I: Configured from console by console
```

Evidence 06: Ping still succeeds with cached ARP

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Evidence 07: ARP cache cleared

```
C:\>arp -a
No ARP Entries Found
```

Evidence 08: Remote ping fails

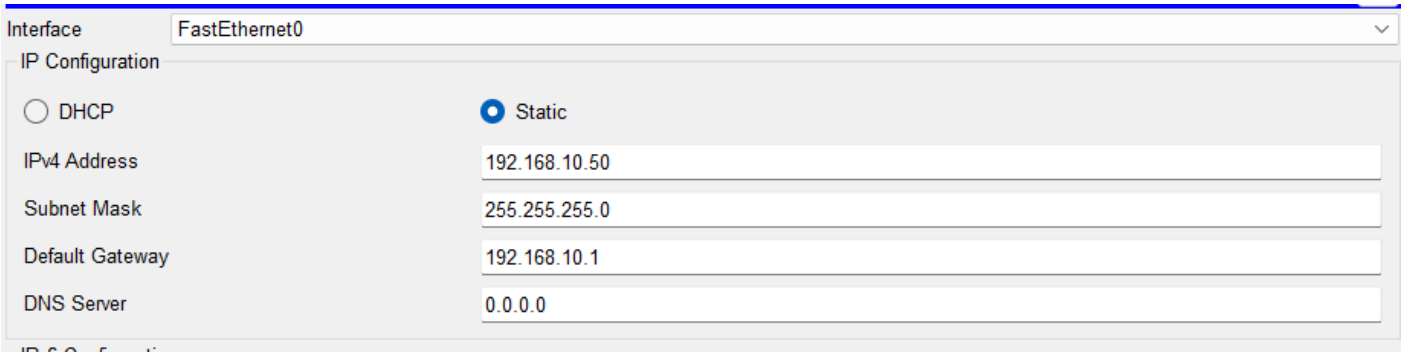
```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Evidence 09: Subnet mask corrected



Evidence 10: Ping succeeds with correct mask

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Evidence 11: Proxy ARP restored

```
Router#show ip interface gigabitEthernet 0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
```

INC-004 - Incorrect DHCP gateway

Evidence 01: Working DHCP baseline

```
Router#  show running-config | section include ip dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp excluded-address 192.168.20.1 192.168.20.9
ip dhcp pool lan10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool LAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
```

Evidence 02: Incorrect DHCP gateway in LAN20 pool

```
Router#  show running-config | section include ip dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp excluded-address 192.168.20.1 192.168.20.9
ip dhcp pool lan10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool LAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.254
 dns-server 8.8.8.8
Router#
```

Evidence 03: Existing client lease shows old gateway

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::2E0:B0FF:FE9B:7698
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.20.10
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.20.1

Bluetooth Connection:
```

Evidence 04: Renewed lease receives incorrect gateway

```
   DNS Server.....................: 0.0.0.0

C:\>ipconfig /renew

   IP Address.....................: 192.168.20.10
   Subnet Mask....................: 255.255.255.0
   Default Gateway................: 192.168.20.254
   DNS Server.....................: 8.8.8.8

C:\>
```

Evidence 05: Remote ping fails

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Request timed out.
Request timed out.

Ping statistics for 192.168.10.1:
    Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
```

Evidence 06: DHCP pool gateway repaired

```
ip dhcp pool LAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
!
```

Evidence 07: Renewed lease receives correct gateway

```
C:\>ipconfig /renew

   IP Address.....................: 192.168.20.10
   Subnet Mask....................: 255.255.255.0
   Default Gateway................: 192.168.20.1
   DNS Server.....................: 8.8.8.8
```

Evidence 08: Remote ping succeeds

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

INC-005 - DHCP network statement

Evidence 01: Router configuration saved

```
Router# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Evidence 02: Incorrect DHCP network definition

```
Router#show running-config | section include ip dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp excluded-address 192.168.20.1 192.168.20.9
ip dhcp pool lan10
 network 0.0.0.0 0.0.0.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool LAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
```

Evidence 03: DHCP request failed

```
C:\>ipconfig /renew
DHCP request failed.

C:\>DHCP request failed.

C:\>DHCP request failed.

C:\>DHCP request failed.

C:\>DHCP request failed.

C:\>DHCP request failed.

C:\>DHCP request failed.
```

Evidence 04: DHCP pool repaired

```
Router#show running-config | section include ip dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp excluded-address 192.168.20.1 192.168.20.9
ip dhcp pool lan10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool LAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
```

Evidence 05: DHCP lease successful

```
C:\>ipconfig /renew

   IP Address.....................: 192.168.10.10
   Subnet Mask....................: 255.255.255.0
   Default Gateway................: 192.168.10.1
   DNS Server.....................: 8.8.8.8
```